NEF ABUNDANCE: FRAMEWORK

1. Theorem and proof

Theorem. *Let X be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor on X such that $\mathrm{n}(X, D) = 3$ and $D^3 = 0$. Then*

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \qquad \textit{for some integer } m > 0.$$

Proof. Suppose for contradiction that $\kappa(D) = -\infty$, equivalently $H^0(X, \mathcal{O}_X(mD)) = 0$ for every integer $m > 0$.

Setup.

- Fix an ample Cartier divisor A and $\epsilon \in \mathbb{Q}_{>0}$. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^3)^{1/3}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

  Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^3 \to \epsilon^3$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large n.

- Let $p_i : X \times X \to X$ be the projections, for $i = 1, 2$. Put

$$Z = \mathbb{P}_{X \times X}\big(p_1^* \mathcal{O}_X(D) \oplus p_2^* \mathcal{O}_X(D)\big).$$

  Let $S_1, S_2 \subset Z$ be the two summand sections and set

$$\mathbb{T} = Z \setminus (S_1 \cup S_2).$$

- Let $D_i, P_{n,i}$ denote the pullbacks of D, P_n by p_i, and also their pullbacks to Z. Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $N > 0$, put

$$R_n = P_{n,1} + P_{n,2} + N\xi.$$

  Whenever $|kR_n|$ is used, $k > 0$ is an integer with kP_n Cartier.

For $x \in X$, let $\mathcal{I}_x$ be the ideal sheaf of the point x. An *exceptional functional* is a linear functional $\Phi : N^1(X)_{\mathbb{R}} \to \mathbb{R}$ supplied with data $(x, \pi : V \to X, H_1, H_2)$, where V is a smooth projective threefold, π is birational, the extended ideal

$$\mathcal{I}_x \mathcal{O}_V := \mathrm{Im}(\pi^* \mathcal{I}_x \to \mathcal{O}_V)$$

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is invertible and equals $\mathcal{O}_V(-E_V)$ for an effective π-exceptional Cartier divisor E_V, and H_1, H_2 are ample Cartier divisors on V. Define

$$\Phi(L) = \frac{\pi^*L \cdot H_1 \cdot H_2}{E_V \cdot H_1 \cdot H_2}.$$

All intersections are on V, and the denominator is positive since $E_V \neq 0$. For every effective Cartier divisor F on X, $\mathrm{mult}_x(F) \leq \Phi(F)$: a local equation of F is a section of $\mathcal{I}_x^{\mathrm{mult}_x(F)}$ near x, so $\pi^*F \geq \mathrm{mult}_x(F)E_V$; intersect with H_1, H_2 and divide by the positive denominator.

For a Cartier divisor L on a smooth projective variety W, $|L|$ *separates s-jets at z* if, for every integer $0 \leq j \leq s$, the natural map

$$\{\sigma \in H^0(W, \mathcal{O}_W(L)) : \sigma_z \in \mathfrak{m}_z^j \mathcal{O}_{W,z}(L)\}$$
$$\longrightarrow \mathfrak{m}_z^j \mathcal{O}_{W,z}(L)/\mathfrak{m}_z^{j+1}\mathcal{O}_{W,z}(L)$$

is surjective, where $s \geq 0$, $\mathcal{O}_{W,z}(L)$ is the stalk, and the map sends a section to the class of its germ. For $s = 1$, this is generation of the value and separation of tangent vectors (Hartshorne, II, Proposition 7.3(2)). We take ϵ sufficiently small that

$$343\epsilon^3 < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

We use the following three lemmas under the above setting. Their proofs are not included in this temporary framework.

Lemma 1.1. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$, $D^3 = 0$ and $\kappa(D) = -\infty$, and let A be an ample Cartier divisor. Fix $\epsilon \in \mathbb{Q}_{>0}$ with $343\epsilon^3 < 1/4$ and $40\epsilon < 3/8$. Let $P_n = r_n(D + A/n)$ be the sequence defined in Setup. For every integer $N > 0$, there exists an integer $n_0 = n_0(\epsilon, N) \geq 1$ such that, for every integer $n \geq n_0$, there exists an exceptional functional Φ with $\Phi(P_n + ND) < 40\epsilon$. The functional and its presentation may depend on n, N; n_0 is independent of these subsequent choices.*

Lemma 1.2. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$ and $D^3 = 0$, and let A be an ample Cartier divisor. Fix $\epsilon \in \mathbb{Q}_{>0}$ with $343\epsilon^3 < 1/4$, and let P_n, R_n be defined as in Setup. For every integer $N > 0$, there exists an integer $n_1 = n_1(\epsilon, N) \geq 1$ such that, for every integer $n \geq n_1$, if there exists an exceptional functional Φ with $\Phi(P_n + ND) < \frac{3}{8}$, then $|kR_n|$ does not separate $8k$-jets at any $z \in \mathbb{T}$, for any integer $k > 0$ with kP_n Cartier.*

For fixed N, take n_1 so that $343P_n^3 < 1/4$ and $r_n > N$ for all $n \geq n_1$, using $P_n^3 \to \epsilon^3$, $343\epsilon^3 < 1/4$ and $r_n \to \infty$. Since $P_n - r_nD$ is nef,

$$(2P_n + ND) \cdot P_n^2 \leq (2 + N/r_n)P_n^3 < 4P_n^3.$$

These are the numerical conditions for the geometric input in Lemma 1.2; its proof is still not included.

Lemma 1.3. *Let X be a smooth projective threefold over $\mathbb{C}$ with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor with $\mathrm{n}(X, D) = 3$ and $D^3 = 0$, and let A be an ample Cartier divisor. Assume $\kappa(D) = -\infty$. Fix $\epsilon \in \mathbb{Q}_{>0}$ with $343\epsilon^3 < 1/4$ and $40\epsilon < 3/8$, and let P_n, Z, R_n be defined as in Setup. There exists an integer $N > 0$ such that, for every $n \gg 0$, $|kR_n|$ separates $8k$-jets at some $z \in \mathbb{T}$ for some integer $k > 0$ with kP_n Cartier.*

We now complete the proof under the standing assumption $\kappa(D) = -\infty$. Lemma 1.1 uses this assumption to produce a small exceptional functional; Lemma 1.2 uses that functional to prohibit jet separation, without requiring $\kappa(D) = -\infty$. Lemma 1.3 uses $\kappa(D) = -\infty$ to provide jet generation. Thus these geometric inputs rule out $\kappa(D) = -\infty$ under the common geometric hypotheses. Choose N by Lemma 1.3. For this fixed N, take $n \geq \max\{n_0(\epsilon, N), n_1(\epsilon, N)\}$ also large enough for Lemma 1.3. By Lemma 1.1, there exists Φ such that

$$\Phi(P_n + ND) < 40\epsilon < \tfrac{3}{8}.$$

Lemma 1.2 then says that $|kR_n|$ does not separate $8k$-jets at any $z \in \mathbb{T}$ for any integer $k > 0$ with kP_n Cartier. But Lemma 1.3, for the same N, n, provides such an integer k and a point $z \in \mathbb{T}$ at which $|kR_n|$ separates $8k$-jets. This is a contradiction. Hence $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for some integer $m > 0$. □

Draft status. This is a reformulation of the first-section argument with its three geometric inputs retained, rather than an independent verification of those inputs. The chosen presentation of Φ, including the model $\pi : V \to X$, the divisors E_V, H_1, H_2, and the extended-ideal identity $\mathcal{I}_x\mathcal{O}_V = \mathcal{O}_V(-E_V)$ must be fixed before reduction of characteristic; the bare linear functional does not supply these data. This AI-assisted draft is for discussion and checking. No claim of novelty, priority, or authorship credit is made.